

Coding & Solution

Date	15 July 2026
Team ID	XXXXXX
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary:

Field	Details
Repository Link / URL	https://github.com/Farisa-11/OptiCrop-Smart-Agricultural-Production-Optimization-Engine
Programming Language(s)	Python 3.x, HTML5, CSS3

Framework(s) Used	Flask, Scikit-learn, Bootstrap
Key Features Implemented	• Crop recommendation based on soil nutrients and environmental conditions. • User-friendly web interface for data input and prediction. • Machine Learning model integration using Flask. • Real-time crop prediction using trained model (.pkl). • Responsive web pages with HTML, CSS, and Bootstrap.
Pending / Incomplete Features	• Weather API integration for real-time climate data. • User authentication and profile management. • Deployment on a cloud platform and multilingual support.
Setup / Run Instructions	1. Clone the GitHub repository. 2. Create and activate a Python virtual environment. 3. Install dependencies using pip install -r requirements.txt. 4. Navigate to the 5. Project Development Phase folder. 5. Run the application using python src/app.py. 6. Open http://127.0.0.1:5000 in a web browser.

Code Quality Checklist:

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

The **OptiCrop** application is developed using **Python**, **Flask**, and **Scikit-learn** to provide intelligent crop recommendations based on soil nutrients and environmental conditions. The project follows a modular architecture with separate folders for datasets, models, source code, templates, and static resources, making it easy to maintain and extend. The trained machine learning model is integrated into a Flask web application, enabling real-time crop prediction through a simple and responsive user interface. The solution is designed to support farmers in making informed crop selection decisions while promoting sustainable agricultural practices.